

2022-2023 春夏学期大气污染控制工程期末考

一、选择题（20 分）

1. 哪个不是温室气体

- A. NO_2 B. CO_2 C. N_2O D. 甲烷

2. 哪个烟羽形状不包括逆温

- A. 爬升型 B. 受限型 C. 蔓烟型 D. 锥形

3. 过滤除尘原理中筛滤作用描述正确的是

4. 生物法适用的什么样的有机废气

- A. 低浓度，高水溶性 B. 高浓度，占地面积大
C. 低浓度，低水溶性 D. 高浓度，……

二、简答题（20 分）

1. 蒸汽和气的共同点和差异性

2. 浆液 pH 和液气比是石灰石-石膏法的重要影响因素，简述为什么要控制在一定范围内。

三、计算题（60 分）

1. 气体吸附，要求用 Ergun 方法计算活性炭床压降，类似下题（10 分）

Example 12.1

An activated-carbon bed that is $12\text{ ft} \times 6\text{ ft} \times 2\text{ ft}$ deep is used in a benzene recovery system. The system is on-line for one hour and is then regenerated for one hour. The influent gas stream contains 5000 ppm benzene (by volume) at 1.0 atm and 100 F. The operating capacity of the bed is 10 lb_m benzene per 100 lb_m carbon. The physical properties of the carbon are as follows: bulk density = $30\text{ lb}_m/\text{ft}^3$, void fraction = 0.40, and particle size = 4×10 mesh (0.011 ft). Determine the pressure drop across the bed from (a) Eq. (12.13), (b) Eq. (12.14), and (c) Figure 12.6.

2. 大气污染排放与扩散，类似中文教材例 10-1（15 分）

给出了稳定度，有效源高，求下风向 2000 米、水平垂直距离 100 米处一个点所增加的 SO_2 浓度及对下风向 3000 米计算点的浓度贡献。

3. 气体吸收，我觉得很难，计算量很大，参考下题（20 分）

Example 13.1

A 10,000 acfm exhaust from a heat treating process contains 25 mole percent ammonia, and the average exhaust conditions are 115 F

and 1.0 atm. Estimate the height of packing, the tower diameter, and the tower pressure drop for an absorption tower that will provide 95% ammonia removal based on the following design parameters:

Average tower temperature is 86 F

Tower pressure is 1.0 atm

Pure water is used as absorbing liquid

Water rate is 1.5 times minimum

Packing is 1.0-in. ceramic Raschig rings

$$k_y a \text{ is } 15 \frac{\text{lbmol}}{\text{hr-ft}^3 \cdot \Delta y}$$

$$k_x a \text{ is } 60 \frac{\text{lbmol}}{\text{hr-ft}^3 \cdot \Delta x}$$

(1) 求最小进液量（单位为 lbm/hr）

(2) 求填料塔的直径和高度（实际进液量为最小进液量的 1.3 倍，在 50%液泛点处操作）

4. VOC 焚烧（15 分）

(1) 利用 Lee 方法计算燃烧温度，去除率为 99%（废气有苯、甲苯等三种物质组成，苯的自燃温度最高），类似下题

Example 11.1

Estimate the temperature required in an isothermal plug flow incinerator with a residence time of 0.5 sec to give 99.5% destruction of toluene. Use the three methods discussed in this section.

(2) 冷却到 200F，采用热回收节省的钱

(3) 有必要安装吗？类似下题

Example 11.4

Calculate the enthalpy change for $500 \text{ m}^3/\text{min}$ of air at 1 atm and 700 C being cooled to 200 C. If fuel gas (methane) costs \$6.50/GJ, and if electricity costs \$0.08/kWh, calculate the daily savings from recovering the heat content of this stream by reducing the temperature to 200 C. Assume that the increased pressure drop due to the heat exchanger is 5 in. H_2O . The existing fan can handle the ΔP but is only 60% efficient. If a heat exchanger to do the job has a TIC of \$100,000 and the appropriate capital recovery factor is 0.18, is the heat exchanger justified? Assume the system operates 250 days per year. If another heat exchanger could be purchased for \$50,000 to reduce the temperature an additional 50 C with an additional pressure drop of 3 in. H_2O , is the second heat exchanger justified?